UNIVERSITY OF PERADENIYA
DEPARTMENT OF PHYSICS

END SEMESTER EXAMINATION
SEMESTER II - 2012/2013

General Physics II - PHY 102

Time allowed: *Two hours*

Answer **FIVE** questions only selecting **TWO** from **Section A**, **TWO** from **Section B** and **ONE** from **Section C**.

(Use of calculators are allowed)

Electron charge (e)	$= 1.60 \times 10^{-19}$ C
Rest mass of electron (m_e)	$= 9.11 \times 10^{-31}$ kg
Permittivity of free space (ε_0)	$= 8.85 \times 10^{-12}$ F m^{-1}
Permeability of free space (μ_0)	$= 4\pi \times 10^{-7}$ H m^{-1}
Planck's constant (h)	$= 6.63 \times 10^{-34}$ J s
Speed of light (c)	$= 3.00 \times 10^{8}$ m s^{-1}
Boltzmann constant (k)	$= 1.38 \times 10^{-23}$ J K^{-1}
Avogadro number (N_A)	$= 6.02 \times 10^{23}$ mol^{-1}.

(Answer Section A in a separate book)

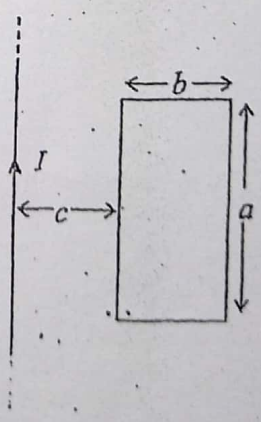
Section-A

(a) Clearly identifying the quantities involved, write down the Ampere's circuital law.

(b) An infinitely long straight wire of radius R carries a uniformly distributed steady current I. Derive expressions for the magnetic flux density at a distance r from the axis of the wire for the cases where $r > R$ and $r \leq R$.

(c) Sketch the variation of the magnetic flux density with r.

(d) Suppose a rectangular conducting loop of length a and width b is placed in a plane through the axis of the infinitely long current carrying straight wire so that the longer side is parallel to the wire and at a distance c from it. Obtain an expression for the magnetic flux through the loop.



(a) With the aid of a clearly labeled diagram, explain the action of a cyclotron.

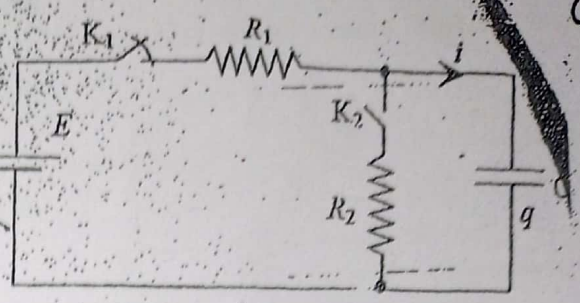
(b) A cyclotron has dees of radius R and magnetic field of B. Obtain an expression for the cyclotron frequency to accelerate ions of charge q and mass m.

(c) If $R = 80$ cm and a magnetic field $B = 1.5$ T find the cyclotron frequency necessary to accelerate deuterons of mass 3.3×10^{-27} kg and charge 1.6×10^{-19} C.

(d) What is the maximum kinetic energy of deuterons coming out from the above cyclotron?

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3. A capacitor of capacitance C is connected to a battery with e.m.f. E and, resistors with resistances R_1 and R_2 as shown in the figure. The key K_2 is initially opened and the key K_1 is closed at time $t = 0$.



(a) Derive an expression for the charge q in the capacitor at any time t.
(b) Obtain an expression for the current i as a function of time t.
(c) If $E = 12$ V, $R_1 = 10$ kΩ, $R_2 = 5$ kΩ and $C = 40$ μF, calculate
(i) the time constant of the circuit and
(ii) charge of the capacitor and the current in the circuit at time $t = 0.1$ s.

(d) After the capacitor is fully charged, the key K_2 is also closed. Now consider the current through R_1 and R_2 as i_1 and i_2 respectively.
(i) What will happen to i_1 and i_2 when K_2 is closed?
(ii) What was value of charge C before closing K_2?
(ii) What will be the value of charge C long time after the key K_2 is closed?

(Answer Sections B and C together in separate books)

SECTION B

4. (a) Use the postulates of Bohr theory to derive an expression for the total energy of the electron in the nth orbit of a singly ionized helium atom.

Given that the binding energy of the electron in a hydrogen atom is 13.6 eV, draw an energy level diagram indicating energies (in eV) of the ground state and the three excited states ($n = 2, 3$ and 4) of the helium atom.

(b) Calculate the wavelength of the possible absorption line involving transition from the ground state to the first excited state.

5. Briefly explain the origin of characteristic (line) X-rays.

An X-ray tube is operated at a peak voltage of 100 kV and the beam current is 40 mA.
(a) What is the power of the tube?
(b) What is the wavelength of the electrons?
(c) How many electrons reach the target per second?
(d) What is the maximum energy of each photon?
(e) What is the maximum momentum of each photon?
(f) What is the minimum (cut-off) wavelength of X-rays?
(g) When X-rays of wavelength 1.54 Å are used to determine the d-spacing (separation between adjacent planes) of a crystal, the Bragg angle for the first order diffraction is 30°. Calculate the d-spacing of the crystal.
(h) When X-rays of wavelength 1.54 Å are allowed to scatter by electrons, the scattered X-rays make 90° to the original direction. What is the wavelength of the scattered X-rays?

6. (a) The following nuclear reaction is responsible for the generation of energy in the sun.

$$4\,{}^{1}_{1}H \rightarrow {}^{4}_{2}He + 2e^{+} + \text{energy}$$

Mass of the H nucleus $= 1.6686 \times 10^{-27}$ kg
Mass of the He nucleus $= 6.6267 \times 10^{-27}$ kg

If the rate of energy output of the sun is 3.9×10^{26} J s^{-1}, calculate the mass of He production per second.

(b) A radiographer has calculated that a patient is to be injected with 1×10^{18} atoms of iodine-131 to monitor the thyroid activity. The half-life of iodine-131 is 8 days. Calculate,

(i) the radioactive decay constant of iodine-131,
(ii) the initial activity of iodine-131,
(iii) the number of undecayed atoms of iodine-131 after 24 days and
(iv) the total activity of iodine-131after 3 days.

SECTION – C

7. (a) Briefly explain what you understand by,

(i) single crystalline materials.
(ii) amorphous materials.
(iii) superconducting materials.
(iv) metallic glasses.
(v) covalent bonding.
(vi) metallic bonding.

(ii) List 5 characteristic properties that make the ceramic materials suitable for heat engine applications. Use the variation of interatomic potential energy with the separation of two atoms to discuss three of these properties.

8. (a) Discuss briefly the rotational and vibrational motion of molecules and indicate the relative energies associated with these two excitations. Draw an energy level diagram showing molecular vibrational and rotational energy levels.

(b) The rotational quantum number $j = 0$ to $j = 1$ transition of a CO molecule occurs at a frequency of 8.15×10^{11} Hz. Calculate the moment of inertia and the bond length of the of the CO molecule.

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